

Nagaveda Sai Kumar Reddy Gajjala

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EDUCATION

University of Texas at Dallas

Master of Science in Computer Science; GPA: 3.94/4.0

Richardson, TX

Jan 2024 - Dec 2025

Karunya Institute of Technology and Sciences

Bachelor of Technology in Computer Science; GPA: 9.6/10

Coimbatore, India

Aug 2018 - May 2022

TECHNICAL SKILLS

Languages: Java, Python, C#, C++, JavaScript, TypeScript, SQL, Bash, HTML5/CSS3

Core Concepts: Data Structures, Algorithms, Object Oriented Design, Operating Systems, Computer Networks, System Design

Frameworks: Spring Boot, .NET Core, ReactJS, Redux, NodeJS, Tailwind CSS, ExpressJS, Hibernate

Cloud & Distributed Systems: AWS, Kafka, Redis, Docker, Kubernetes, CI/CD, Microservices

Machine Learning (AI) & Data: LangChain, RAGs, Pandas, NumPy, Scikit-Learn

Databases & Tools: PostgreSQL, MySQL, MongoDB, Git, Linux, Maven

WORK EXPERIENCE

Paycom

Irving, TX

Software Developer Intern (Full Stack)

May 2025 – Aug 2025

- Designed and built a Full Stack Vendor Communication Portal (React, .NET Core) by engineering **reusable Higher-Order Components (HOCs)** and custom authentication utilities to facilitate secure interactions, replacing manual email workflows with **LDAP-enforced** access control.
- Designed a fault-tolerant polling service using **Cron jobs** to sync external email replies with the internal database, maintaining real-time consistency across a hierarchical message thread model (Events → Threads → Messages).
- Implemented Granular **Role-Based Access Control (RBAC)** to govern conversation visibility and vendor onboarding workflows, ensuring strict data isolation between vendor entities.
- Won the Paycom Codeathon (2025): Secured **1st place** for developing a QR based swag collection tracker prototype, recognized for technical innovation and business impact.

University of Texas at Dallas(On Campus)

Richardson, TX

Software Developer

May 2024 – Dec 2025

- Led the migration of massive **monolithic** 3rd-party enterprise codebases from **Java 8 to 17**, resolving **60+** complex dependency conflicts and leveraging modern Garbage Collection (GC) to improve performance for client-facing systems.
- Modernized the build infrastructure for large-scale **legacy monoliths** by “Mavenizing” the codebase, standardizing dependency management and reducing CI/CD build times to ensure reliable delivery for external stakeholders.

Zoho Corporation (Global SaaS)

Chennai, India

Software Developer (EMS Framework Backend)

Dec 2021 – Nov 2023

- Architected a deterministic state machine to handle **64+ complex onboarding scenarios** (8x8 matrix) for the *Endpoint Central & Service Desk Plus* integration, reconciling cross-product “Waiting for Approval” (WFA) security gates and license states.
- Engineered a fault-tolerant bi-directional synchronization engine with **paginated batch processing** to recover failed transactions, ensuring **eventual consistency** and supporting asynchronous migration pipelines for legacy data upgrades.
- Optimized high-throughput REST APIs (Jersey) for the **EMS Framework**, utilizing **Redis** caching (Look-aside) and **database indexing** strategies to slash average response latency by **65%**.
- Implemented **Kafka**-based queues to decouple high-volume device ingestion from the onboarding business logic, enabling the system to scale to massive concurrent requests without blocking.
- Built internal release management tools that automated workflows, reducing release cycles by **50%** and improving product stability in a fast-paced **Agile/Scrum** environment.

PROJECTS

Ride-Sharing Backend (Uber Clone) | *Spring Boot, Kafka, Redis, PostGIS*

- Designed a scalable backend using **Spring Boot** and **Apache Kafka** to handle high-throughput driver location updates, implementing **Geospatial Indexing (PostGIS)** for sub-10ms driver matching queries.
- Implemented the **Strategy Design Pattern** to dynamically switch pricing algorithms (surge vs. standard) based on real-time demand metrics stored in **Redis**.

Intelligent Doc-Search Engine (RAG Pipeline) | *Python, LangChain, Pinecone, Redis*

- Engineered a **Retrieval-Augmented Generation (RAG)** pipeline to query large technical documentation, leveraging **LangChain** for orchestration and **Pinecone** for high-dimensional vector storage.
- Implemented **Hybrid Search** (Semantic + Keyword) and re-ranking algorithms to improve context retrieval accuracy by **40%**, reducing LLM hallucinations on domain-specific queries.
- Optimized query latency by implementing **semantic caching** with **Redis**, storing frequently asked embedding vectors to bypass redundant API calls and reduce token costs.

Monolith to Microservices Refactoring | *Java, Docker, Kubernetes, gRPC*

- Refactored a monolithic application into a distributed **Microservices architecture**, decoupling user authentication and feed services using **Docker** and **Kubernetes** for independent scaling.
- Established inter-service communication using high-performance **gRPC** (Protobufs) for low-latency synchronous calls and asynchronous event messaging for data consistency.